

Game-Theoretic Fuzzy Reinforcement Learning via Self-Triggered Mechanism

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Abstract—This paper proposes a game-theoretic fuzzy reinforcement learning framework for multi-player continuous-time nonzero-sum differential games, aiming to approximate the Nash equilibrium strategies. To avoid the continuous condition checking required by conventional event-triggered schemes, we develop a self-triggered update mechanism, where each player refreshes its feedback input only at scheduled triggering instants using intermittent state measurements. The coupled Hamilton–Jacobi equations are approximated by fuzzy logic systems embedded in a hybrid Actor–Critic architecture: the Critics evolve continuously, while the Actors are updated only at triggering instants. To improve data efficiency and relax the dependence on persistent excitation, an experience replay mechanism is incorporated into the learning laws. Moreover, a projection-based update is adopted to keep the Actor weights within prescribed bounds, mitigating parameter drift and enhancing learning safety. Rigorous theoretical analysis and simulation studies demonstrate the effectiveness of the proposed learning algorithm.

Index Terms—Fuzzy reinforcement learning, nonzero-sum game, Actor-Critic learning, self-triggered mechanism.

I. INTRODUCTION

Multi-player nonzero-sum games (NZSGs) have attracted considerable attention for modeling strategic interactions in engineering systems such as multi-agent coordination [1], robotic networks [2], and unmanned vehicles [3], where multiple decision-makers pursue individual yet coupled objectives [4]. Unlike zero-sum games, NZSG formulations are suitable for these scenarios because the objectives among agents are interactive but not strictly antagonistic [4], [5]. However, computing Nash equilibrium strategies remains difficult due to the coupled Hamilton-Jacobi (HJ) equations [6]–[8]. For

complex nonlinear engineering systems, reinforcement learning (RL) and adaptive dynamic programming (ADP) provide an online way to approximate value functions and policy mappings directly [9]–[13]. Motivated by fuzzy modeling and RL, this paper employs fuzzy logic systems (FLSs) [14], [15] in an Actor-Critic RL framework to approximate both value functions and Nash policies, thereby developing adaptive fuzzy learning laws for NZSGs.

Related Work

RL with FLS-based function approximation provides an effective framework for synthesizing equilibrium strategies, yet its continuous-time implementation typically requires continuous sensing and frequent policy/state updates [16]. Such requirements are difficult to meet in resource-constrained digital platforms, where sensing, computation, and communication budgets are limited. To alleviate this burden, intermittent update mechanisms—particularly event-triggered control (ETC) [17]—have received extensive attention. By executing aperiodic updates only when a state-dependent triggering condition is violated, ETC can significantly reduce energy and communication usage [2], [18]–[20]. It has also been combined with optimal control, consensus protocols, and RL architectures [3], [21], [22], illustrating broad applicability. Nevertheless, ETC still hinges on continuous state monitoring to evaluate the triggering condition, which may diminish its practical advantages and increase hardware complexity [23]. To avoid continuous monitoring, self-triggered control (STC) has emerged as a proactive alternative [24]. In STC, future update instants are predicted using sampled information and system evolution, so that sensing can be performed only at scheduled times [25]–[27]. This feature aligns naturally with fuzzy RL algorithms, where one seeks data-efficient learning with structured and interpretable function approximators. Motivated by this, the present work incorporates STC into adaptive fuzzy RL for multi-player NZSGs, aiming to reduce sampling frequency while ensuring stable learning and bounded approximation of Nash strategies.

While the STC mechanism provides an effective framework for reducing sampling and sensing efforts within the interaction of NZSG players, guaranteeing stable learning under sparse policy/state updates is still nontrivial. In particular, for Actor–Critic RL algorithms, approximation errors and intermittent information may cause parameter drift, and the resulting uncontrolled growth of weight estimates can trigger divergence during online learning. To mitigate this issue,

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projected update mechanisms are commonly employed to constrain adaptive parameters within user-specified compact sets during weight adaptation [28]. The projection operator (PO) not only guarantees bounded learning trajectories but also maintains desirable convergence characteristics under standard RL algorithms [29]. This safeguard is especially important in continuous-time ADP/RL, where large gradients may otherwise destabilize the learning dynamics [9].

In addition to boundedness guarantees, the learning efficacy of Actor–Critic RL also depends critically on the informativeness of collected data. Many continuous-time RL/ADP schemes impose the persistency of excitation (PE) condition [11] to ensure parameter convergence, which can be demanding in practice [30]. To relax this requirement, the experience replay (ER) mechanism has become a widely used, sample-efficient approach to improve learning richness and support convergence. By storing and reutilizing past data, ER increases the diversity of learning signals without raising the online sampling rate [11]. Moreover, historical transitions can be exploited to reconstruct Bellman residuals or learning errors, enabling more stable and efficient learning from finite data [30], [31]. Motivated by these insights, a projected and ER-enhanced adaptive fuzzy RL framework is developed for NZSGs under self-triggered execution.

Contribution: The main contributions of this article are summarized as follows.

- 1) Continuous condition checking in ETC can impose a non-negligible sensing/computation burden [3], [17]. To eliminate this requirement, a self-triggered mechanism is developed to explicitly determine future update instants, thereby reducing online computation while preserving stability and control performance.
- 2) Unconstrained weight adaptation may suffer from parameter drift and even destabilize the control input, especially under sparse updates [32], [33]. The proposed projected learning algorithm keeps the Actor weights within prescribed bounds, preventing divergence and improving learning safety.
- 3) Convergence within Actor-Critic RL under intermittent feedback often depends on PE condition, which is difficult to guarantee in practice and can degrade system performance [27], [29]. A data-efficient ER mechanism is incorporated with the RL algorithm to reuse informative historical data, enrich learning signals under limited sampling, and support uniform ultimate boundedness (UUB) of the Actor-Critic weight errors.

Structure: The remainder of this article is arranged as follows. Section II provides the mathematical preliminaries of the FLSs and the projected gradient descent (GD). In Section III, we present the problem of the NZSGs, followed by the triggering framework in Section IV. The fuzzy learning algorithm is proposed in Section V with theoretical justifications in Section VI. Simulation studies are conducted in Section VII, followed by the conclusions in Section VIII.

Notation: The abbreviations “w.r.t.”, “s.t.”, and “iff” are used throughout this paper to refer to “with respect to”, “such that”, and “if and only if” respectively. The notations listed in Table I are adopted in this paper.

TABLE I
NOTATIONS

Notation	Description
$\triangleq$	equality by definition
$\mathbb{N}$	the set of all natural numbers
$\mathbb{R}^n$	the n -dimensional Euclidean space
$\mathbb{R}^{m \times n}$	the set of real matrices of dimension $m \times n$
C^1	the set of functions with continuous partial derivatives
Ω	a compact (i.e. bounded and closed) subset of $\mathbb{R}^n$
$A^\top$	the transpose of matrix A
$\ x\ $	$\triangleq \sqrt{x^\top x}$, the Euclidean norm of $x \in \mathbb{R}^n$
$\bar{\lambda}(A)$	the maximum eigenvalue of matrix A
$\underline{\lambda}(A)$	the minimum eigenvalue of matrix A
$A \succ 0$	A is positive definite and symmetric
$f \succ 0$	f is continuous and positive definite
Δf	the difference of function f
∇f	the gradient of function f
$\nabla^\top f$	the transpose of ∇f
$\mathbf{1}\{\Phi\}$	the indicator function: $= 1$ if Φ is true, and 0 otherwise
$\mathbf{N}$	$\triangleq \{1, 2, \dots, N\}$, the set of player indices
a_{-i}	$\triangleq \{a_j : j \in \mathbf{N} \text{ and } j \neq i\}$ for a finite sequence $\langle a_i \rangle_{i=1}^N$
$\sum_i, \sum_j$	abbreviations of $\sum_{i=1}^N$ and $\sum_{j=1}^N$, respectively.

II. PRELIMINARIES

A. Fuzzy Logic Systems

Definition 1. [14] A fuzzy logic system (FLS) is defined as a 4-tuple $\{\mathcal{F}, \mathcal{I}, \mathcal{D}, \mathcal{R}\}$ with $\mathcal{F}$ being the fuzzifier, $\mathcal{I}$ the fuzzy inference engine based on fuzzy rules, $\mathcal{D}$ the defuzzifier, and $\mathcal{R}$ the knowledge base composed of P fuzzy IF–THEN inference rules, where the p -th rule is given by

$$\begin{aligned} \text{IF} : x_1 \in X_1^p \wedge x_2 \in X_2^p \wedge \dots \wedge x_n \in X_n^p \\ \text{THEN} : y \in Y^p \quad \forall p = 1, 2, \dots, P, \end{aligned}$$

with $x_i \in \mathbb{R}$ being the input, $y \in \mathbb{R}$ the output, and X_i^p, Y^p the fuzzy sets.

Accordingly, through the membership functions $\mu_{X_i^p}$ and μ_{Y^p} , the output y of the FLS can be reformulated as

$$y(x) = \frac{\sum_{p=1}^P \max_{y \in \mathbb{R}} \mu_{Y^p}(y) \prod_{i=1}^n \mu_{X_i^p}(x_i)}{\sum_{p=1}^P \prod_{i=1}^n \mu_{X_i^p}(x_i)} \quad (1)$$

where $x = [x_1, \dots, x_n]^\top$. Alternatively, the output of FLS (1) can be given by a compact form as $y(x) = W^\top \varphi(x)$, where

$$\begin{aligned} W &= [\max_{y \in \mathbb{R}} \mu_{Y^1}(y), \dots, \max_{y \in \mathbb{R}} \mu_{Y^P}(y)]^\top, \\ \varphi(x) &= [\varphi_1(x), \dots, \varphi_P(x)]^\top, \end{aligned}$$

and the fuzzy basis function (FBF) is given by

$$\varphi_p(x) = \prod_{i=1}^n \mu_{X_i^p}(x_i) / \sum_{p=1}^P \prod_{i=1}^n \mu_{X_i^p}(x_i).$$

The FLS can be used for universal function approximation, as summarized in the following lemma.

Lemma 1. [15] For a real-valued function $f(x)$ defined on a compact set Ω , for given $\varepsilon > 0$, there exists a FLS in the form of $y(x) = W^\top \varphi(x)$ s.t. $\sup_{x \in \Omega} |f(x) - y(x)| \leq \varepsilon$.

B. Projected Gradient Descent

Definition 2 (Projection Operator [34]). For $\omega \in \mathbb{R}^n$ and $\sigma > 0$, the PO is defined as

$$\mathcal{P}(\omega | \sigma) \triangleq \omega + \mathbf{1}\{\sigma < \|\omega\|\} \cdot \frac{\sigma - \|\omega\|}{\|\omega\|} \omega$$

The discrete-time projected GD based on the PO and its boundedness are given as follows.

Lemma 2. [28] Consider the minimization problem

$$\min_{\omega} J(\omega) \quad \text{subject to } \omega \in \Phi \triangleq \{\omega \in \mathbb{R}^n \mid \|\omega\| \leq \sigma\} \quad (2)$$

w.r.t. given convex objective function $J \in C^1$ and $\sigma > 0$, with its ideal solution $\omega^* \triangleq \arg \min_{\omega \in \Phi} J(\omega) \in \Phi$. Then, under the projected GD law with sufficiently small $\alpha > 0$:

$$\omega_{k+1} = \mathcal{P}(\bar{\omega}_k | \sigma), \quad \bar{\omega}_k = \omega_k - \alpha \nabla J(\omega_k), \quad \omega_0 \in \Phi,$$

(i) ω_k remains in Φ , $\forall k \in \mathbb{N}$; (ii) $\lim_{k \rightarrow \infty} \omega_k = \omega^*$; (iii) $(\bar{\omega}_k - \omega^*)^\top (\bar{\omega}_k - \bar{\omega}_k) \geq 0$, $\forall k \in \mathbb{N}$ and $\bar{\omega}_k \triangleq \sigma \cdot \bar{\omega}_k / \|\bar{\omega}_k\|$.

III. PROBLEM STATEMENT

Consider the continuous-time N -player nonlinear system

$$\dot{x} = f(x) + \sum_i g_i(x) u_i, \quad x(0) = x_0 \in \Omega, \quad (3)$$

where $x \in \mathbb{R}^n$, $\Omega \subset \mathbb{R}^n$ is compact and contains a neighborhood of the origin, and u_i is the input of player $i \in \mathbb{N}$. The drift f and input dynamics g_i are Lipschitz continuous on Ω , with $f(0) = 0$, and satisfy $\|f(x)\| \leq \mathcal{L}_f \|x\|$ and $\|g_i(x)\| \leq \mathcal{G}_i$, $\forall i \in \mathbb{N}$, for some $\mathcal{L}_f > 0$ and $\mathcal{G}_i > 0$.

The system (3) is assumed stabilizable on Ω , which is a standard assumption in game-RL for infinite-horizon differential games, e.g., see [8], [31]. See Table I for the other notations not explained here and after.

To evaluate the control performance, the corresponding cost function w.r.t. the system (3) of each player i is defined as

$$J_i(u_i, u_{-i}; x_0) = \int_0^\infty c_i(x(\tau), u_i(\tau), u_{-i}(\tau)) d\tau, \quad (4)$$

$$c_i(x, u_i, u_{-i}) \triangleq Q_i(x) + \sum_j r_{ij} u_j^2,$$

where $c_i(x, u_i, u_{-i})$ is the instantaneous cost with $Q_i(\cdot) \succ 0$ and constants $r_{ij} \geq 0$ and $r_{ii} > 0$. Here, $Q_i(\cdot)$ by definition has a quadratic lower bound with some $\underline{q}_i > 0$ s.t.

$$Q_i(x) \geq \underline{q}_i \|x\|^2 \quad \forall x \in \Omega. \quad (5)$$

Definition 3 (Admissible Control Tuple, [7]). On the compact set $\Omega \subset \mathbb{R}^n$, the feedback policies $u_i \equiv u_i(x)$ of all players are said to constitute an admissible tuple, denoted by $\{u_i\}_{i=1}^N \in \mathbf{U}(\Omega)$, if each u_i is continuous with $u_i(0) = 0$, each cost (4) is finite for all $x_0 \in \Omega$, and it asymptotically stabilizes (3).

For the system (3) under admissible control tuple $\{u_i\}_{i=1}^N$, $Q_i(x) \succ 0$ implies that the cost $c_i(x, u_i, u_{-i})$ vanishes only at the origin, which further ensures the zero-state observability (ZSO) of the system (3) [31], [35].

The value function associated with player i is defined as

$$V_i(x(t); u_i, u_{-i}) \triangleq \int_t^\infty c_i(x(\tau); u_i, u_{-i}) d\tau, \quad (6)$$

where $c_i(x; u_i, u_{-i})$ abbreviates $c_i(x, u_i(x), u_{-i}(x))$ for simplicity. On the compact set Ω , we suppose that V_i is continuously differentiable [35], i.e., $V_i \in C^1$ and thus uniquely solves the corresponding Bellman equation: $V_i(0) = 0$ and

$$\nabla^\top V_i(\cdot; u_i, u_{-i}) \cdot (f + \sum_j g_j u_j) + c_i(\cdot; u_i, u_{-i}) = 0. \quad (7)$$

Since the system (3) has ZSO guaranteed by $Q(x) \succ 0$, V_i is positive definite for all $i \in \mathbb{N}$ and can serve as a Lyapunov function in the subsequent analysis [7], [35].

The control objective of each player i is to minimize its value function (6), subject to $\{u_i\}_{i=1}^N \in \mathbf{U}(\Omega)$. Therefore, based on the game theory [4], the multi-player NZSG aims to find an admissible tuple $\{u_i^*\}_{i=1}^N$ that achieves a Nash equilibrium defined below.

Definition 4 (Nash Equilibrium [4]). An admissible tuple $\{u_i^*\}_{i=1}^N$ constitutes a Nash equilibrium to the N -player NZSG if for any $i \in \mathbb{N}$ and policy u_i s.t. $\{u_i, u_{-i}^*\} \in \mathbf{U}(\Omega)$,

$$V_i(x; u_i^*, u_{-i}^*) \leq V_i(x; u_i, u_{-i}^*) \quad \forall x \in \Omega. \quad (8)$$

For simplicity, we write $V_i^*(x) \triangleq V_i(x; u_i^*, u_{-i}^*)$; similarly, $c_i^*(x) \triangleq c_i(x; u_i^*, u_{-i}^*)$. Enforcing the stationarity condition on the i -th Hamiltonian $\mathcal{H}_i^*$ w.r.t. V_i^* and c_i^* :

$\mathcal{H}_i^*(x, u_i, u_{-i}) \triangleq c_i^*(x) + \nabla^\top V_i^*(x) (f(x) + \sum_j g_j(x) u_j(x))$, the corresponding Nash policies are given by

$$\frac{\partial \mathcal{H}_i^*}{\partial u_i} = 0 \quad \Rightarrow \quad u_i^*(x) = -\frac{1}{2r_{ii}} g_i^\top(x) \nabla V_i^*(x). \quad (9)$$

For well-definedness and technical reasons, each Nash policy u_i^* is assumed Lipschitz continuous over Ω , i.e., $\exists \mathcal{L}_i > 0$ s.t.

$$|u_i^*(x) - u_i^*(y)| \leq \mathcal{L}_i \|x - y\| \quad \forall x, y \in \Omega. \quad (10)$$

Furthermore, since $V_i^* \in C^1$, ∇V_i^* is bounded on compact set Ω ($\cdot$ is continuous), i.e., there exists $\delta_i > 0$ s.t.

$$\sup_{x \in \Omega} \|\nabla V_i^*(x)\| \leq \delta_i. \quad (11)$$

Substituting (9) into (7) then yields the following N -coupled HJ equations [5], [7]:

$$0 = \sum_j \frac{r_{ij} (g_j^\top \nabla V_j^*)^2}{4r_{jj}^2} + Q_i + \nabla^\top V_i^* (f - \sum_j \frac{g_j g_j^\top \nabla V_j^*}{2r_{jj}}).$$

The Nash strategy (9) requires continuous state measurements, incurring substantial computation and inter-player communication. To reduce this burden, a sample-based mechanism is developed under the following standard assumptions.

Assumption 1. [24] For system (3) with an admissible tuple $\{u_i\}_{i=1}^N$, $\exists K > 0$ s.t. for all $x, y \in \Omega$, and $z \triangleq x - y$:

$$\begin{aligned} x^\top [f(x) + \sum_i g_i(x) u_i(y)] &\leq K(1 + \|x\|^2 + \|y\|^2), \\ z^\top [f(x) + \sum_i g_i(x) u_i(y)] &\leq K(1 + \|x\|^2 + \|y\|^2). \end{aligned}$$

Assumption 2. [36] There exists class $\mathcal{K}_\infty$ functions $\underline{\alpha}_i$ and $\bar{\alpha}_i$, s.t. $\underline{\alpha}_i(\|x\|) \leq V_i^*(x) \leq \bar{\alpha}_i(\|x\|)$, $\forall x \in \Omega$.

IV. SAMPLING RULE DESIGN

A. Event-triggered framework

Denoting $\langle t_k \rangle_{k=1}^\infty$ as an increasing sequence of sample time instants, we employ the intermittent feedback strategy by incorporating an aperiodic sampling mechanism by which the sampled state $\tilde{x}$ is updated as

$$\tilde{x}(t) = x(t_k) \quad \forall t \in [t_k, t_{k+1}), \quad (12)$$

with the sample error $\tilde{x}(t) \triangleq x(t) - \tilde{x}(t)$. Accordingly, each Nash control input with intermittent feedback is given by

$$\tilde{u}_i^*(t) \equiv u_i^*(\tilde{x}) = -\frac{1}{2r_{ii}} g_i^\top(\tilde{x}) \nabla V_i^*(\tilde{x}). \quad (13)$$

Then, under the sample-based control inputs (13), the property of the system (3) can be summarized in the following lemma.

Lemma 3. [33] *Let Assumption 2 hold. Consider the control inputs in (13) with the sampling rule in (12). With $\beta \in (0, 1)$, under the event-triggering condition*

$$\|\tilde{x}\|^2 \leq \frac{2\beta \sum_i q_i}{\sum_i \sum_j \delta_i \mathcal{L}_j^2} \|x\|^2, \quad (14)$$

the state of system (3) is UUB.

B. Self-triggered framework

The conventional event-triggering condition (14) requires continuous state monitoring, resulting in additional computational overhead. To eliminate this requirement, a self-triggered mechanism is introduced to predict trigger instants in advance. The following lemma first establishes the relationships among the state $x(t)$, sampling error $\tilde{x}(t)$, and sampled state $\tilde{x}(t)$.

Lemma 4. [23] *Let Assumption 1 hold. Under the sampling process (12), the sample error $\tilde{x}(t)$ and the state $x(t)$ admit the following upper and lower bounds, respectively:*

$$\|\tilde{x}(t)\|^2 \leq \Upsilon_1(\|\tilde{x}\|, t - t_k), \quad (15)$$

$$\|x(t)\|^2 \geq \Upsilon_2(\|\tilde{x}\|, t - t_k), \quad (16)$$

where

$$\begin{aligned} \Upsilon_1(\|\tilde{x}\|, t) &\triangleq \frac{2\|\tilde{x}\|^2 + 1}{3} (e^{6Kt} - 1), \\ \Upsilon_2(\|\tilde{x}\|, t) &\triangleq \frac{5\|\tilde{x}\|^2 + 1}{3} e^{-6Kt} - \frac{2\|\tilde{x}\|^2 + 1}{3}. \end{aligned}$$

Proof. The proof can be found in [24, Lemma 2]. $\square$

Remark 1. *Assumption 1 is a standard growth condition in self-triggered sampled-data analysis [23], [24]. It requires the closed-loop dynamics under an admissible strategy tuple to grow at most quadratically on Ω , without imposing structural restrictions on (3). This mild condition for locally Lipschitz systems ensures forward completeness [36] and is used to derive the subsequent self-triggered sampling rule.*

The following theorem then formulates a Zeno-free self-triggered sampling rule based on the bounds (15) and (16).

Theorem 1. *Under the sampling process (12), for given positive parameters η_1 and η_2 , a class $\mathcal{K}$ function $\Psi_1(\cdot)$, and*

a class $\mathcal{K}_\infty$ function $\Psi_2(\cdot)$, if the self-triggered mechanism establishes the following inequality:

$$\Psi_2(\Upsilon_1(\|\tilde{x}\|, t - t_k)) \leq \eta_1 \Psi_1(\Upsilon_2(\|\tilde{x}\|, t - t_k) + \eta_2), \quad (17)$$

then the Zeno phenomenon is excluded.

Proof. Let $\chi \triangleq \|\tilde{x}\|$ for notational brevity. First, we show the existence and uniqueness of the solution to the threshold equation established as $\Theta(t; t_k) = 0$, where

$$\Theta(t; t_k) \triangleq \Psi_2(\Upsilon_1(\chi, t - t_k)) - \eta_1 \Psi_1(\Upsilon_2(\chi, t - t_k) + \eta_2).$$

Notice that $\Upsilon_1(r, t)$ and $\Upsilon_2(r, t)$ are increasing and decreasing in t , respectively. Recall (17) and the class $\mathcal{K}$ functions Ψ_1, Ψ_2 . Then, the threshold function $\Theta(t; t_k)$ is clearly monotonically increasing in t . Combining it with the facts that $\Theta(t_k; t_k) < 0$ and $\lim_{t \rightarrow \infty} \Theta(t; t_k) > 0$, we can conclude that the solution to $\Theta(t; t_k) = 0$ uniquely exists on $[t_k, +\infty)$, which can be considered as the triggering instant t_{k+1} next to t_k .

Then, the self-triggered instant determined by solving $\Theta(t; t_k) = 0$ is ensured to exclude the Zeno behavior by showing that $\Delta t_k = t_{k+1} - t_k$ is lower-bounded by a positive state-dependent variable. To this end, one can always find a continuous function $\Psi_3(\cdot)$ s.t. $0 < \Psi_3(\chi) < \chi^2 + \eta_2$ for $\chi \geq 0$. Inspired by [24], to verify the strict positiveness of the lower bound of Δt_k , consider a sufficient condition

$$\Upsilon_2(\chi, \Delta t_k) + \eta_2 \leq \Psi_3(\chi) \leq \Upsilon_1(\chi, \Delta t_k).$$

Then, one can obtain $\Delta t_k \geq \tau_k \triangleq \frac{1}{6K} \min\{\Gamma_1(\chi), \Gamma_2(\chi)\}$ with $\Gamma_1(\chi) \triangleq \ln\left(1 + \frac{3\Psi_3(\chi)}{1+2\chi^2}\right)$ and $\Gamma_2(\chi) \triangleq -\ln\left(1 - \frac{3(\chi^2 + \eta_2 - \Psi_3(\chi))}{5\chi^2 + 1}\right)$. It is straightforward that $\tau_k > 0$ since $\Gamma_1(\chi) > 0$ and $\Gamma_2(\chi) > 0$. Even if the state converges to the equilibrium s.t. $\chi \rightarrow 0$, $\tau_k > 0$ also holds by

$$\lim_{\chi \rightarrow 0} \Gamma_1(\chi) = \ln(1 + 3\Psi_3(0)) > 0,$$

$$\lim_{\chi \rightarrow 0} \Gamma_2(\chi) = -\ln(1 - 3(\eta_2 - \Psi_3(0))) > 0,$$

since $0 < \Psi_3(0)$. Thus, the Zeno behavior is excluded. $\square$

We now turn our attention to the stability of the system (3) by establishing the relation between $\|\tilde{x}\|$ and $\|x\|$ through the self-triggering sampling mechanism (17).

Lemma 5. *Let the functions Ψ_1 and Ψ_2 in Theorem 1 satisfy*

$$\Psi_1(\chi) \leq \chi \leq \Psi_2(\chi) \quad \forall \chi \geq 0. \quad (18)$$

Then, under Assumption 1,

$$\|\tilde{x}\|^2 \leq \eta_1 \|x\|^2 + \eta_1 \eta_2. \quad (19)$$

Proof. Since Ψ_1 and Ψ_2 are class $\mathcal{K}$, inserting (15) and (16) into (17) yields $\Psi_2(\|\tilde{x}\|^2) \leq \eta_1 \Psi_1(\|x\|^2 + \eta_2)$. Hence,

$$\|\tilde{x}\|^2 \leq \Psi_2(\|\tilde{x}\|^2) \leq \eta_1 \Psi_1(\|x\|^2 + \eta_2) \leq \eta_1 \|x\|^2 + \eta_1 \eta_2,$$

by (18), which completes the proof. $\square$

In the ETC setting, one implements the condition (14) to achieve UUB property but leads to computational overhead. Under the self-triggering condition (17), we replace the terms $\|\tilde{x}\|^2$ and $\|x\|^2$ in (14) by the inequalities (15) and (16) in

Lemma 4, which eliminates continuous state measurement, as detailed in the following theorem.

Theorem 2. *Let Assumptions 1 and 2 hold, and the functions Ψ_1 and Ψ_2 in the self-triggering rule (17) further satisfy (18) with sufficiently small $\eta_1, \eta_2 > 0$. Then, under the sample-based Nash equilibrium (13), the system state $x(t)$ is UUB.*

Proof. See Appendix A. $\square$

Remark 2. *In the self-triggered mechanism (17), the class $\mathcal{K}$ functions Ψ_1 and Ψ_2 should be determined to satisfy (18). For simplicity, one can choose $\Psi_1(\chi) = \chi$ and $\Psi_2(\chi) = \chi + p\chi^2$ for $p \geq 0$. Moreover, as discussed in Theorem 2, smaller η_1 and η_2 result in a tighter bound for the system state x .*

Remark 3. *In Theorem 1, the execution instant is determined by solving the scalar threshold equation*

$$\Theta(t; t_k) = 0. \quad (20)$$

Although (20) is transcendental in t , suitable variable substitution and the class- $\mathcal{K}$ functions in Remark 2 reduce it to a quadratic algebraic equation with low computational cost [23]. Since it is solved only at each triggering instant t_k , its overhead is negligible compared with the continuous control and learning computations.

V. HYBRID FUZZY ACTOR-CRITIC RL

Since solving the coupled nonlinear HJ equations for the Nash value functions V_i^* is generally intractable, a learning-based ADP scheme is developed to approximate V_i^* and the sampled policies (13). As shown in Figure 1, each player minimizes its value function under self-triggered intermittent feedback using a fuzzy Actor-Critic RL framework with POs. Composite learning further reuses historical data to achieve sufficient excitation (SE), thereby relaxing the PE requirement.

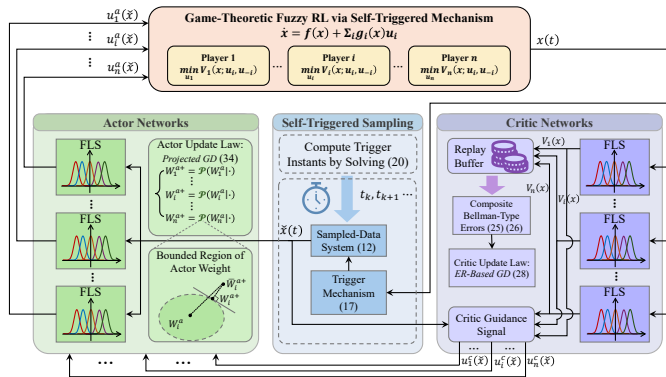


Fig. 1. Illustration of the proposed methodology, where W_i^a denotes the Actor weight, $V_i(x)$ the (approximate) value function, $u_i^a(\tilde{x})$ the input with intermittent feedback, and $u_i^c(\tilde{x})$ the guidance signal of Actor.

Definition 5. [37] *A bounded vector signal $w(t)$ is said to be:*
(i) *PE*, if $\exists \underline{w}, T > 0$ s.t. $\int_t^{t+T} w(\tau)w^\top(\tau) d\tau \geq \underline{w}I \forall t \geq 0$;
(ii) *SE over a finite horizon*, if $\exists \underline{w}, T > 0$ and $t_j \in (0, T)$ for $j \in \{1, 2, \dots, M\}$ s.t. $\sum_{j=1}^M w(t_j)w^\top(t_j) \geq \underline{w}I$.

A. Function Approximation with FLSs

With the FLSs discussed in Definition 1 and Lemma 1, the value functions $\{V_i^*\}_{i \in \mathbf{N}}$ at the Nash equilibrium can be expressed by the ideal Critic neural networks (NNs) as

$$V_i^*(x) = W_i^{c* \top} \psi_i^c(x) + \varepsilon_i^c(x) \quad (21)$$

where $W_i^{c*} \in \mathbb{R}^{N_i^c}$ is the ideal Critic weights, $\psi_i^c : \mathbb{R}^n \rightarrow \mathbb{R}^{N_i^c}$ is the basis function, and $\varepsilon_i^c : \mathbb{R}^n \rightarrow \mathbb{R}$ is the parameterization error, with N_i^c being the number of the nodes of the Critic NN, for Player $i \in \mathbf{N}$. Moreover, each Nash control input (9) can be parameterized by the Actor NN as

$$u_i^*(x) = W_i^{a* \top} \psi_i^a(x) + \varepsilon_i^a(x) \quad \text{for } i \in \mathbf{N}, \quad (22)$$

with the ideal Actor weight $W_i^{a*} \in \mathbb{R}^{N_i^a}$, the basis function $\psi_i^a : \mathbb{R}^n \rightarrow \mathbb{R}^{N_i^a}$, and the residual error $\varepsilon_i^a : \mathbb{R}^n \rightarrow \mathbb{R}$, where N_i^a is the number of the i -th Actor nodes. These function approximations satisfy the following standard assumption.

Assumption 3. [35] *The basis $\psi_i^s(\cdot)$ and the residual $\varepsilon_i^s(\cdot)$ are bounded over the compact set Ω as*

$$\|\psi_i^s(\cdot)\| \leq \bar{\psi}^s, \|\nabla \psi_i^s(\cdot)\| \leq \bar{\psi}_d^s, \|\varepsilon_i^s(\cdot)\| \leq \bar{\varepsilon}^s, \|\nabla \varepsilon_i^s(\cdot)\| \leq \bar{\varepsilon}_d^s \quad \text{by positive constants } \bar{\psi}^s, \bar{\psi}_d^s, \bar{\varepsilon}^s, \bar{\varepsilon}_d^s, \text{ and } s \in \{a, c\}.$$

Since the ideal weights W_i^{c*} and W_i^{a*} remain unknown, we define the adaptive Critic $V_i(\cdot)$ and Actor $u_i^a(\cdot)$ as

$$V_i(x) = W_i^{c \top} \psi_i^c(x), \quad u_i^a(x) = W_i^{a \top} \psi_i^a(x) \quad (23)$$

with their respective online adaptive weights W_i^c and W_i^a , and with a slight abuse of notation. The associated weight errors are then defined as $\tilde{W}_i^c \triangleq W_i^{c*} - W_i^c$ and $\tilde{W}_i^a \triangleq W_i^{a*} - W_i^a$.

B. Continuous Critic Learning

In what follows, we will interchangeably write

$F, F(x), F(x(t))$ and $F(t)$ (resp. $\tilde{F}, \tilde{F}(\tilde{x}), \tilde{F}(\tilde{x}(t))$ and $\tilde{F}(t)$)

for an arbitrary function F of state x (resp. of its sample $\tilde{x}$) if there is no confusion, for notational simplicity. E.g., we denote ε_i^a and ψ_i^a with the self-triggered sampling as $\tilde{\varepsilon}_i^a(t) \triangleq \varepsilon_i^a(\tilde{x}(t))$ and $\tilde{\psi}_i^a(t) \triangleq \psi_i^a(\tilde{x}(t))$. The control inputs generated by the adaptive Actor NNs (23) with sampled state feedback are then given by $\tilde{u}_i^a(t) \triangleq u_i^a(\tilde{x}(t)) = W_i^{a \top}(t) \tilde{\psi}_i^a(t)$, under which the Critic learning error at time t is defined as

$$e_i^c(t) \triangleq \mathcal{H}_i(t) - \mathcal{H}_i^*(t) = W_i^{c \top}(t) \cdot \varpi_i(t) + c_i(t), \quad (24)$$

$$\text{where } \begin{cases} \varpi_i(t) \triangleq \nabla \psi_i^c(t) \cdot (f(t) + \sum_j g_j(t) \tilde{u}_j^a(t)) \\ c_i(t) \triangleq c_i(x(t); \tilde{u}_i^a(t), \tilde{u}_{-i}^a(t)) \\ \mathcal{H}_i(t) \triangleq c_i(t) + \nabla^\top V_i(t) \cdot (f(t) + \sum_j g_j(t) \tilde{u}_j^a(t)) \\ \mathcal{H}_i^*(t) \triangleq \mathcal{H}_i^*(x(t), u_i^*(t), u_{-i}^*(t)) = 0 \end{cases}$$

In order to relax the PE condition, consider the composite Critic error at time t w.r.t. the data at past instants $t_{\ell}^c < t$

$$e_{i\ell}^c(t) \triangleq W_i^{c \top}(t) \cdot \varpi_i(\tau) + c_i(\tau) \Big|_{\tau=t_{\ell}^c} \quad (25)$$

for $\ell = 1, \dots, L_i^c$. Let $\varpi_{i\ell} \triangleq \varpi_i(t_{i\ell}^c)$ for brevity. Then, each Critic learning aims to minimize the objective

$$E_i^c \triangleq \frac{1}{2} \left(\frac{e_i^c}{1 + \|\varpi_{i\ell}\|^2} \right)^2 + \frac{1}{2} \sum_{\ell=1}^{L_i^c} \left(\frac{e_{i\ell}^c}{1 + \|\varpi_{i\ell}\|^2} \right)^2 \quad (26)$$

via each normalized GD

$$\dot{W}_i^c = -\alpha_i^c \left(\frac{\varpi_{i\ell} e_i^c}{(1 + \|\varpi_{i\ell}\|^2)^2} + \sum_{\ell=1}^{L_i^c} \frac{\varpi_{i\ell} e_{i\ell}^c}{(1 + \|\varpi_{i\ell}\|^2)^2} \right), \quad (27)$$

where $\alpha_i^c > 0$ is the learning rate for each Critic learning.

For implementation, the historical data in (25) are stored in a finite replay buffer with L_i^c samples for player i . After $\{\varpi_{i1}, \dots, \varpi_{iL_i^c}\}$ contains N_i^c linearly independent vectors, no new samples are added, and the summations in (26) and (27) are evaluated over this fixed memory set [30].

C. Projected Actor Learning

In (22), each Nash control input (9) is represented by the Actor NN directly. As an alternative, given the Critic NN (21), each Nash control input (9) can also be expressed as

$$u_i^*(x) = -\frac{1}{2r_{ii}} g_i^\top(x) \cdot [\nabla \psi_i^{c\top}(x) W_i^{c*} + \nabla \varepsilon_i^c(x)]. \quad (28)$$

Then, define the guidance of Actor learner

$$\check{u}_i^c(t) \triangleq -\frac{1}{2r_{ii}} \check{g}_i^\top(t) \cdot \nabla \check{\psi}_i^{c\top}(t) \cdot W_i^c(t), \quad (29)$$

with self-triggered sampling, and each Actor's learning error w.r.t. the guidance (29)

$$e_i^a \triangleq \check{u}_i^a - \check{u}_i^c = W_i^{a\top} \check{\psi}_i^a + \frac{1}{2r_{ii}} \check{g}_i^\top \nabla \check{\psi}_i^{c\top} W_i^c = -\tilde{W}_i^{a\top} \check{\psi}_i^a + \zeta_i^a$$

where $\zeta_i^a \triangleq W_i^{a\top} \check{\psi}_i^a + \frac{1}{2r_{ii}} \check{g}_i^\top \nabla \check{\psi}_i^{c\top} W_i^c$.

It should be noted that (22) is adopted as the implemented Actor because it directly parameterizes the policy by an independent Actor network. By contrast, (28) is a Critic-induced representation and is then used only to construct the guidance signal in (29), rather than as the implemented policy.

Similarly to the Critic case, define the composite Actor learning error at the past instants $t_{i\ell}^a < t$ for $\ell = 1, \dots, L_i^a$ as

$$e_{i\ell}^a(t) \triangleq -\tilde{W}_i^{a\top}(t) \cdot \check{\psi}_i^a(\tau) + \zeta_i^a(\tau) \Big|_{\tau=t_{i\ell}^a} \quad (30)$$

and let $\check{\psi}_{i\ell}^a \triangleq \check{\psi}_i^a(t_{i\ell}^a)$ and $\zeta_{i\ell}^a \triangleq \zeta_i^a(t_{i\ell}^a)$. The target function to be minimized by each Actor is

$$E_i^a \triangleq \frac{1}{2} (e_i^a)^2 + \frac{1}{2} \sum_{\ell=1}^{L_i^a} (e_{i\ell}^a)^2 \quad (31)$$

Then, the projected GD law is proposed as

$$\begin{aligned} \dot{W}_i^a(t) &= 0 & \text{for } t \in [t_k, t_{k+1}) \\ W_i^{a+}(t) &= \mathcal{P}(\bar{W}_i^a(t) | \rho_i) & \text{for } t = t_{k+1}, \end{aligned} \quad (32)$$

where ρ_i is the user-defined constraint on the adaptive weight of each Actor, $\bar{W}_i^a \triangleq W_i^a - \alpha_i^a (\check{\psi}_i^a e_i^a + \sum_{\ell=1}^{L_i^a} \check{\psi}_{i\ell}^a e_{i\ell}^a)$, and $\alpha_i^a > 0$ is the Actor learning rate. Here, ρ_i defines the compact feasible set $\Omega_i^a \triangleq \{W_i^a \in \mathbb{R}^{N_i^a} : |W_i^a| \leq \rho_i\}$ and should conservatively ensure $W_i^{a*} \in \Omega_i^a$. Increasing ρ_i enlarges the search region and weakens projection, whereas an overly small value strengthens the constraint but may cause projection bias.

In the following discussions, we employ the Actor version (22) as the parameterized Nash control input (9).

VI. STABILITY ANALYSIS OF HYBRID SYSTEM

In this subsection, we conduct the stability analysis of the system (3) in a hybrid manner. Specifically, the state and adaptive weight dynamics are discussed in Section VI-A and Section VI-B, respectively.

A. Stability of the State Feedback Dynamics

1) **Continuous Dynamics of State** $x(t)$: With the sample-based control inputs $\check{u}_i^a(t)$, the dynamics (3) is represented as

$$\dot{x}(t) = f(t) + \sum_j g_j(t) \check{u}_j^a(t). \quad (33)$$

For its stability analysis, consider $V_i^*(x)$ as the Lyapunov candidate. Differentiating $V_i^*(x)$ along (33) then yields

$$\begin{aligned} \dot{V}_i^* &= \nabla^\top V_i^* \cdot (f + \sum_j g_j \check{u}_j^a) \\ &= \nabla^\top V_i^* \cdot \sum_j g_j (\check{u}_j^a - \check{u}_j^*) + \nabla^\top V_i^* \cdot (f + \sum_j g_j \check{u}_j^*), \end{aligned}$$

where the term $\nabla^\top V_i^* \cdot \sum_j g_j (\check{u}_j^a - \check{u}_j^*)$ can be bounded using (11) and $\|g_j\| \leq \mathcal{G}_j$ as

$$\begin{aligned} \nabla^\top V_i^* \cdot \sum_j g_j (\check{u}_j^a - \check{u}_j^*) &\leq \|\nabla^\top V_i^*\| \cdot \sum_j |g_j| \cdot |\check{u}_j^a - \check{u}_j^*| \\ &\leq \delta_i \sum_j (\mathcal{G}_j \cdot |\check{u}_j^a - \check{u}_j^*|). \end{aligned} \quad (34)$$

Moreover, the term $|\check{u}_j^a - \check{u}_j^*|$ can be further constrained as

$$\begin{aligned} |\check{u}_j^a - \check{u}_j^*| &= |W_j^{a\top} \check{\psi}_j^a - W_j^{a* \top} \check{\psi}_j^a - \check{\varepsilon}_j^a| \\ &\leq |W_j^{a\top} \check{\psi}_j^a| + |W_j^{a* \top} \check{\psi}_j^a + \check{\varepsilon}_j^a|, \end{aligned}$$

where W_j^a is bounded as $\|W_j^a\| \leq \rho_j$ by the Actor learning (32) and Lemma 2; Assumption 3 further bounds $|\check{u}_j^a - \check{u}_j^*|$ as

$$|\check{u}_j^a - \check{u}_j^*| \leq \rho_j \cdot \bar{\psi}^a + \|W_j^{a*}\| \cdot \bar{\psi}^a + \bar{\varepsilon}^a. \quad (35)$$

Then, following the proof of Theorem 2 in Appendix A, the term $\nabla^\top V_i^*(t) \cdot (f(t) + \sum_j g_j(t) \check{u}_j^*(t))$ is upper bounded by

$$\nabla^\top V_i^*(t) (f(t) + \sum_j g_j(t) \check{u}_j^*(t)) \leq -q_i \|x(t)\|^2 + \delta_i \mathcal{G}_L \|\tilde{x}(t)\|$$

for $\mathcal{G}_L \triangleq \sum_j \mathcal{G}_j \mathcal{L}_j$. Therefore, with above inequality, $\dot{V}_i^*(x)$ is bounded by the above results including (34) and (35) as

$$\dot{V}_i^*(t) \leq -q_i \|x(t)\|^2 + \delta_i \cdot \mathcal{G}_L \|\tilde{x}(t)\| + \delta_i \cdot \mathcal{E} \quad (36)$$

where $\mathcal{E} \triangleq \sum_j \mathcal{G}_j \cdot (\rho_j \bar{\psi}^a + \|W_j^{a*}\| \cdot \bar{\psi}^a + \bar{\varepsilon}^a)$. It can be seen from (35) that ρ_i directly affects the bound of $|\check{u}_j^a - \check{u}_j^*|$, and consequently influences the constant $\mathcal{E}$ in (36), which affects the bound on $\|x\|$ as discussed subsequently.

2) **Dynamics of the Sampled State** $\check{x}(t)$: Under the sampling process (12), the jump dynamics of $\check{x}(t)$ satisfies

$$\begin{aligned} \dot{\check{x}}(t) &= 0 & \text{for } t \in [t_k, t_{k+1}), \\ \check{x}^+(t) &= x(t) & \text{for } t = t_{k+1}. \end{aligned} \quad (37)$$

Built upon the dynamics in (33) and (37), the following analysis establishes the UUB property of both continuous and sampled states under the self-triggered sampling.

Theorem 3. *Let Assumptions 1 to 3 hold. Under the state sampling process (12) and the self-triggering rule (17) with sufficiently small η_1, η_2 , consider the control inputs of the form*

(23) with the Actor-Critic RL tuning laws in (27) and (32), the system state $x(t)$ is UUB.

Proof. See Appendix B. $\square$

Corollary 1. Consider the same conditions in Theorem 3. In addition to Assumption 2, suppose there exist positive constants $\underline{\theta}$ and $\bar{\theta}$ s.t.

$$\underline{\theta}\|x\|^2 \leq V(x) \triangleq \sum_i V_i^*(x) \leq \bar{\theta}\|x\|^2. \quad (38)$$

Then, $\tilde{x}(t)$ is UUB satisfying $\|\tilde{x}(t_k)\| \leq \bar{x}$, $k \geq N_{\bar{x}}$ for some constants $N_{\bar{x}} \in \mathbb{N}$ and $\bar{x} > 0$.

Proof. See Appendix B. $\square$

Remark 4. The parameters η_1 and η_2 trade off learning accuracy and sampling frequency. Smaller values tighten the sampling-error bound and increase triggering frequency, whereas larger values reduce sampling at the cost of a larger ultimate bound in (53); see Lemma 5 and Appendix B.

Remark 5. Let r_Ω denote a radius associated with the compact domain Ω . From the derivations in the proofs of Theorems 2 and 3, particularly (50) in Appendix A and (52) in Appendix B, a sufficient condition on the self-triggered parameters η_1 and η_2 is given by:

$$0 < \eta_1 < \frac{(\varrho q)^2 r_\Omega^4}{(\delta \mathcal{G}_L)^2 (r_\Omega^2 + \eta_2)}, \quad \varrho \in \{\beta, \kappa_2\}$$

where β , κ_2 , q , δ , and $\mathcal{G}_L$ are defined in the appendices.

B. Stability of the Actor-Critic Weight Dynamics

1) **Dynamics of the Actor weight error** $\tilde{W}_i^a(t)$: From the Actor weight update law, its discrete error dynamics can be derived as $\dot{\tilde{W}}_i^a(t) = 0$ for all $t \in [t_k, t_{k+1})$. At the transition time t_{k+1} , the weight error is updated as

$$\begin{aligned} \tilde{W}_i^{a+} &= \tilde{W}_i^{a*} - W_i^{a+} = \tilde{W}_i^{a*} - \mathcal{P}(\tilde{W}_i^a | \rho_i) \\ &= \tilde{W}_i^a - \alpha_i^a \cdot (\tilde{\psi}_i^a \tilde{\psi}_i^{a\top} + \mathcal{A}_{i1}) \tilde{W}_i^a \\ &\quad + \alpha_i^a \mathcal{A}_{i2} - \mathcal{A}_{i3} + \mathcal{A}_{i4} - \alpha_i^a \tilde{\psi}_i^a \tilde{\psi}_i^{a\top} W_i^{a*}. \end{aligned} \quad (39)$$

Here, for technical reasons, the notation below is introduced:

$$\bar{W}_i^{a*} \triangleq W_i^{a*} - \alpha_i^a \tilde{\psi}_i^a (\tilde{u}_i^* - \tilde{u}_i^{c*}) \equiv W_i^{a*},$$

where $\tilde{u}_i^*(t)$ is represented in terms of the ideal Critic NNs as

$$\tilde{u}_i^{c*}(t) \triangleq -\frac{1}{2r_{ii}} \tilde{g}_i^\top(t) \cdot [\nabla \tilde{\psi}_i^{c\top}(t) W_i^{c*} + \nabla \tilde{\varepsilon}_i^c(t)].$$

The remaining “ $\mathcal{A}$ -terms” in (39) are defined as

$$\begin{aligned} \mathcal{A}_{i1} &\triangleq \sum_{\ell=1}^{L_i} \tilde{\psi}_{i\ell}^a \tilde{\psi}_{i\ell}^{a\top}, \quad \mathcal{A}_{i2} \triangleq \tilde{\psi}_i^a \zeta_i^a + \sum_{\ell=1}^{L_i} \tilde{\psi}_{i\ell}^a \zeta_{i\ell}^a \\ \mathcal{A}_{i3} &\triangleq \mathbf{1}\{\rho_i < \|\bar{W}_i^a\|\} \frac{\rho_i - \|\bar{W}_i^a\|}{\|\bar{W}_i^a\|} \bar{W}_i^a \\ \mathcal{A}_{i4} &\triangleq -\frac{1}{2} \alpha_i^a \tilde{\psi}_i^a (r_{ii}^{-1} \tilde{g}_i^\top \nabla \tilde{\psi}_i^{c\top} W_i^{c*} + r_{ii}^{-1} \tilde{g}_i^\top \nabla \tilde{\varepsilon}_i^c \\ &\quad + 2\tilde{\varepsilon}_i^a + 2\tilde{\psi}_i^a W_i^{a*}). \end{aligned} \quad (40)$$

Consider $V_i^a \triangleq \|\tilde{W}_i^a\|^2$ as a Lyapunov candidate for each Actor weight error and its difference $\Delta V_i^a = \|\tilde{W}_i^{a+}\|^2 - \|\tilde{W}_i^a\|^2$. At transition time t_{k+1} , ΔV_i^a can be expressed with (39) as

$$\begin{aligned} \Delta V_i^a &= (\alpha_i^a)^2 \tilde{W}_i^{a\top} (\tilde{\psi}_i^a \tilde{\psi}_i^{a\top} + \mathcal{A}_{i1})^2 \tilde{W}_i^a + \tilde{W}_i^{a\top} \Xi_{i2} \\ &\quad - 2\alpha_i^a \tilde{W}_i^{a\top} (\tilde{\psi}_i^a \tilde{\psi}_i^{a\top} + \mathcal{A}_{i1}) \tilde{W}_i^a + \Xi_{i1} + \Xi_{i3}, \end{aligned} \quad (41)$$

where

$$\begin{aligned} \Xi_{i1} &\triangleq (\alpha_i^a)^2 \|\mathcal{A}_{i2}\|^2 + \|\mathcal{A}_{i4}\|^2 + 2\alpha_i^a \mathcal{A}_{i2}^\top \mathcal{A}_{i4} \\ \Xi_{i2} &\triangleq 2(\alpha_i^a \mathcal{A}_{i2} + \mathcal{A}_{i4}) - 2(\alpha_i^a)^2 (\tilde{\psi}_i^a \tilde{\psi}_i^{a\top} + \mathcal{A}_{i1}) \mathcal{A}_{i2} \\ &\quad - 2\alpha_i^a (\tilde{\psi}_i^a \tilde{\psi}_i^{a\top} + \mathcal{A}_{i1}) \mathcal{A}_{i4} \\ \Xi_{i3} &\triangleq \|\mathcal{A}_{i3}\|^2 - 2\mathcal{A}_{i3}^\top (\bar{W}_i^{a*} - \bar{W}_i^a). \end{aligned} \quad (42)$$

2) **Dynamics of the Critic weight error** $\tilde{W}_i^c(t)$: Recall (24). With the Actor and Critic NNs and $\tilde{u}_i^a(t) \triangleq \tilde{W}_i^{a\top} \tilde{\psi}_i^a + \tilde{\varepsilon}_i^a$, the Hamiltonian terms in the Critic error (24) can be expressed as

$$\begin{aligned} \mathcal{H}_i &= W_i^{c\top} \varpi_i + Q_i + \sum_j r_{ij} \cdot (\tilde{u}_j^a)^2 \\ \mathcal{H}_i^* &= W_i^{c* \top} \nabla \psi_i^c [f + \sum_j g_j \tilde{u}_j^a] + Q_i \\ &\quad + W_i^{c* \top} \nabla \psi_i^c \sum_j g_j [u_j^* - \tilde{u}_j^* + \tilde{u}_j^a] \\ &\quad + \nabla \varepsilon_i^c [f + \sum_j g_j u_j^*] + \sum_j r_{ij} \cdot (u_j^*)^2, \end{aligned}$$

from which one can rewrite the Critic error as

$$e_i^c(t) = -\tilde{W}_i^{c\top}(t) \cdot \varpi_i(t) + \zeta_i^c(t) \quad (43)$$

with $\zeta_i^c(t) \triangleq -\nabla \varepsilon_i^c(t) \cdot f(t) + \sum_{h=1}^3 \gamma_{ih}(t)$ and

$$\begin{aligned} \gamma_{i1} &\triangleq \sum_j r_{ij} \cdot ((\tilde{u}_j^a)^2 - (u_j^*)^2), \quad \gamma_{i2} \triangleq -\nabla \varepsilon_i^c \sum_j g_j u_j^* \\ \gamma_{i3} &\triangleq -W_i^{c\top} \nabla \psi_i^c \sum_j g_j (x)(u_j^* - \tilde{u}_j^* + \tilde{u}_j^a). \end{aligned} \quad (44)$$

Accordingly, one has $e_{i\ell}^c = -\tilde{W}_i^{c\top} \varpi_{i\ell} + \zeta_{i\ell}^c$, where $\zeta_{i\ell}^c \triangleq \zeta_i^c(t_{i\ell})$. Then, the Critic weight error dynamics is given by

$$\dot{\tilde{W}}_i^c(t) = -\alpha_i^c [(\tilde{\varpi}_i \tilde{\varpi}_i^\top + C_{i1}) \tilde{W}_i^c - C_{i2}], \quad (45)$$

where $\tilde{\varpi}_i \triangleq \varpi_i / (1 + \|\varpi_i\|^2)$, $\tilde{\varpi}_{i\ell} \triangleq \varpi_{i\ell} / (1 + \|\varpi_{i\ell}\|^2)$ and

$$\begin{aligned} C_{i1} &\triangleq \sum_{\ell=1}^{L_i} \tilde{\varpi}_{i\ell} \tilde{\varpi}_{i\ell}^\top \\ C_{i2} &\triangleq \frac{\varpi_i \zeta_i^c}{(1 + \|\varpi_i\|^2)^2} + \sum_{\ell=1}^{L_i} \frac{\varpi_{i\ell} \zeta_{i\ell}^c}{(1 + \|\varpi_{i\ell}\|^2)^2}. \end{aligned}$$

Let $V_i^c \triangleq \|\tilde{W}_i^c\|^2 / (2\alpha_i^c)$ be the Lyapunov candidate. Differentiating V_i^c along the dynamics (45) yields

$$\dot{V}_i^c = -\tilde{W}_i^{c\top} \tilde{\varpi}_i \tilde{\varpi}_i^\top \tilde{W}_i^c - \tilde{W}_i^{c\top} C_{i1} \tilde{W}_i^c + \tilde{W}_i^{c\top} C_{i2}, \quad (46)$$

and the stability of the Actor-Critic RL are as follows.

Theorem 4. Let all the conditions in Theorem 3 hold. Then, the Actor-Critic weight errors $\tilde{W}_i^a$ and $\tilde{W}_i^c$ are UUB, provided that the signals $\varpi_{i\ell}$ and $\tilde{\psi}_{i\ell}^a$ are SE for all $i \in \mathbb{N}$ and that $\alpha_i^a > 0$ is sufficiently small.

Proof. See Appendix C. $\square$

Having established the UUB properties of both the system trajectories and the Actor-Critic weight errors separately, we now consider the entire hybrid system that incorporates both continuous and discrete dynamics in the following corollary.

Corollary 2. Under the same conditions, the augmented hybrid state $\mathbf{X} \triangleq [x^\top \tilde{x}^\top \tilde{W}^{a\top} \tilde{W}^{c\top}]^\top \in \mathbb{R}^{2n+\sum_i N_i^a+\sum_i N_i^c}$ is UUB, where $\tilde{W}^a \triangleq [\tilde{W}_1^{a\top} \dots \tilde{W}_N^{a\top}]^\top$ and $\tilde{W}^c \triangleq [\tilde{W}_1^{c\top} \dots \tilde{W}_N^{c\top}]^\top$.

Proof. Let $\mathbf{0}$ be the zero vector in $\mathbb{R}^N$, and construct a hybrid system from (33), (37), (39), and (45) as follows.

(a) Continuous dynamics:

$$\dot{\mathbf{X}}(t) = [\dot{x}^\top(t) \ 0 \ \mathbf{0}^\top \ \dot{\tilde{W}}^{c\top}(t)]^\top$$

during each execution interval $[t_k, t_{k+1})$.

(b) Discrete dynamics: at each trigger instant $t = t_{k+1}$,

$$\mathbf{X}^+ = \mathbf{X}(t_k) + [0 \ \tilde{x}^\top(t_k) \ \mathcal{A}^\top(t_k) \ \mathbf{0}^\top]^\top,$$

where $\mathcal{A} \triangleq [\mathcal{A}_1 \dots \mathcal{A}_N]^\top$, with $\mathcal{A}_i \triangleq -\alpha_i^a(\tilde{\psi}_i^a \tilde{\psi}_i^{a\top} + \mathcal{A}_{i1})\tilde{W}_i^a + \alpha_i^a \mathcal{A}_{i2} - \mathcal{A}_{i3} + \mathcal{A}_{i4} - \alpha_i^a \tilde{\psi}_i^a \tilde{\psi}_i^{a\top} \tilde{W}_i^{a*}$ for each $i \in \mathbf{N}$.

For the stability analysis of the hybrid system above, consider the following Lyapunov candidate

$$\mathbf{V}(\mathbf{X}) = V(x) + V(\tilde{x}) + \sum_i [V_i^a(\tilde{W}_i^a) + V_i^c(\tilde{W}_i^c)],$$

where $V = \sum_i V_i^*$ (see Appendix A).

(a) **Continuous Dynamics:** Since the sampled state and the Actor weights are updated only at the trigger instants and remain unchanged throughout the execution interval, one has

$$\dot{V}(\tilde{x}) = 0 \text{ and } \dot{V}_i^a(\tilde{W}_i^a) = 0$$

for $t \in [t_k, t_{k+1})$. Then, differentiating $\mathbf{V}(\mathbf{X})$ yields

$$\dot{\mathbf{V}}(\mathbf{X}) = \dot{V}(x) + \sum_i \dot{V}_i^c(\tilde{W}_i^c).$$

(b) **Discrete Dynamics:** At the triggering instant $t = t_{k+1}$, the difference of $\mathbf{V}(\mathbf{X})$ can be expressed as

$$\Delta \mathbf{V}(\mathbf{X}) = \mathbf{V}(\mathbf{X}^+) - \mathbf{V}(\mathbf{X}) = \sum_i \Delta V_i^a(\tilde{W}_i^a) + \Delta V(\tilde{x}).$$

Therefore, following the discussions in Theorems 3 and 4 (see Appendices B and C), we can conclude that the augmented hybrid state $\mathbf{X}(t)$ is UUB. $\square$

VII. SIMULATION

To demonstrate the effectiveness of the proposed self-triggered fuzzy RL algorithm, we consider a modified torsional pendulum system [38] as shown in Figure 2 and modeled as

$$\begin{aligned} \frac{dy}{dt} &= \mathbf{w} + \mathbf{y} \cdot u_1, \\ \mathbf{J} \frac{d\mathbf{w}}{dt} &= u_2 - \mathbf{Mgl} \sin \mathbf{y} - \mathbf{p} \frac{dy}{dt} + \mathbf{w} \cdot u_3 + \mathbf{J} \cdot d(t). \end{aligned} \quad (47)$$

where $d(t) = \mathbf{1}\{t \in [20, 23]\} \cdot 0.5$ is an unknown external disturbance, and the parameters in (47) are specified in Table II.

Let $x_1 = \mathbf{y}$, $x_2 = \mathbf{w}$, and $x = [x_1, x_2]^\top$. Then, the perturbed torsional pendulum system (47) can be rewritten as

$$\dot{x} = f(x) + g_1(x)u_1 + g_2(x)u_2 + g_3(x)u_3 + D(t), \quad (48)$$

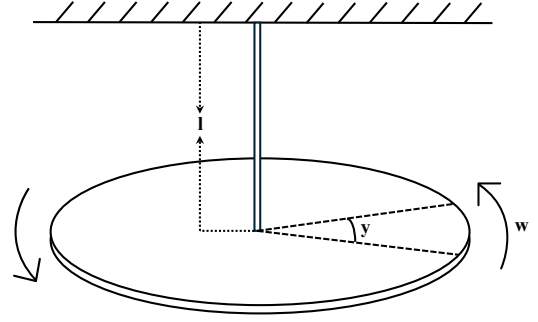


Fig. 2. The modified torsional pendulum.

TABLE II
SYMBOL DEFINITIONS AND VALUES

Parameter	Meaning	Value	Unit
u_1	angular velocity modulation gain	—	rad/s
u_2	primary control torque	—	N · m
u_3	angle-dependent torque input	—	N · m
$\mathbf{y}$	angle	—	rad
$\mathbf{w}$	angular velocity	—	rad/s
$\mathbf{M}$	mass	1/3	kg
l	length of the pendulum bar	2/3	m
$\mathbf{J}$	rotary inertia	$4/3\mathbf{M}l^2$	kg · m ²
$\mathbf{p}$	friction factor	0.2	—
$\mathbf{g}$	gravity acceleration	9.8	m/s ²

where we consider $u_i (i = 1, 2, 3)$ as the control inputs, $D(t) = [0, d(t)]^\top$ is an external disturbance to verify the robustness of the proposed method, and

$$\begin{aligned} f(x) &= [x_2, -(\mathbf{Mgl}/\mathbf{J}) \sin x_1 - (\mathbf{p}/\mathbf{J})x_2]^\top, \\ g_1(x) &= [x_1, -(\mathbf{p}/\mathbf{J})x_1]^\top, g_2(x) = [0, 1/\mathbf{J}]^\top, g_3(x) = [0, x_2/\mathbf{J}]^\top. \end{aligned}$$

The simulation parameters are given in Table III. For all players, the finite replay buffer sizes were set as $L_i^c = L_i^a = 30$. The initial state was $x(0) = [0.012, 0.66]^\top$, and all Actor and Critic weights were independently initialized from $\mathcal{U}[-1, 1]$ under the same initialization rule. In addition, the functions $Q_i(x)$ in (4), $\Psi_1(\chi)$ and $\Psi_2(\chi)$ in (17) are determined as $Q_1(x) = \|x\|^2$, $Q_2(x) = 2\|x\|^2$, $Q_3(x) = 1.5\|x\|^2$, $\Psi_1(\chi) = \chi$ and $\Psi_2(\chi) = \chi$, respectively.

In the simulation, Gaussian membership functions (GMFs) are used as the FBFs of the FLS, with centers and variances fixed *a priori* to cover the operating region. Specifically, the FBFs for Actors and Critics are determined as

$$\begin{aligned} \psi_{a,i}^{\iota,j}(\tilde{x}_1, \tilde{x}_2) &= \frac{\phi_{X_1^i}(\tilde{x}_1) \cdot \phi_{X_2^j}(\tilde{x}_2)}{\sum_{r=1}^5 \sum_{s=1}^5 \phi_{X_1^r}(\tilde{x}_1) \cdot \phi_{X_2^s}(\tilde{x}_2)}, \\ \psi_{c,i}^{\iota,j}(x_1, x_2) &= \frac{\phi_{X_1^i}(x_1) \cdot \phi_{X_2^j}(x_2)}{\sum_{r=1}^5 \sum_{s=1}^5 \phi_{X_1^r}(x_1) \cdot \phi_{X_2^s}(x_2)}, \end{aligned}$$

with $\iota, j = 1, 2, 3, 4, 5$, and the membership functions

$$\phi_{X_1^i}(x_1) = e^{-\frac{(x_1-9+3i)^2}{3}}, \phi_{X_2^j}(x_2) = e^{-\frac{(x_2-9+3j)^2}{3}}.$$

Then, $\psi_{a,i} = [\psi_{a,i}^{1,1}, \dots, \psi_{a,i}^{1,5}, \psi_{a,i}^{2,1}, \dots, \psi_{a,i}^{5,5}]^\top \in \mathbb{R}^{25}$ and $\psi_{c,i} = [\psi_{c,i}^{1,1}, \dots, \psi_{c,i}^{1,5}, \psi_{c,i}^{2,1}, \dots, \psi_{c,i}^{5,5}]^\top \in \mathbb{R}^{25}$.

TABLE III
SIMULATION SETTING

Parameters	r_{1j}	r_{2j}	r_{3j}	α_1^a	α_2^a	α_3^a	α_1^c	α_2^c	α_3^c	ρ_1	ρ_2	ρ_3	$\mathcal{L}_1$	$\mathcal{L}_2$	$\mathcal{L}_3$	K	η_1	η_2
Value	2	1	1.2	0.2	0.3	0.3	0.2	0.3	0.3	1.1	1.2	1.3	1	2	3	0.05	0.43	0.28

A. Comparative studies with event-triggered approach

First, the characteristics of the proposed self-triggered mechanism are analyzed by comparison studies under the same simulation settings. In Table IV, compared with the event-triggered method in [33], the proposed self-triggered scheme requires much fewer triggering instants (51 versus 356) and achieves much larger average and minimal triggering intervals. In particular, the minimal interval of the proposed method is 0.7885s, which is strictly greater than zero. Therefore, the results not only verify that the proposed method excludes Zeno behavior as shown in Theorem 1, but also demonstrate its advantage in reducing triggering frequency.

TABLE IV
COMPARATIVE RESULTS.

Strategies	Metrics	Samples	Average interval	Minimal interval
Proposed method		51	0.8048s	0.7885s
The method in [33]		356	0.1975s	0.0002733s

B. Sensitivity analysis of the self-triggered parameters

Table V reports the effect of the self-triggering parameters η_1 and η_2 . It can be seen that decreasing either η_1 or η_2 generally increases the number of samples, while improving the regulation performance. In particular, the smallest final state norm is achieved under the stricter setting $(0.5\eta_1, 0.5\eta_2)$, at the cost of the largest sampling number. By contrast, enlarging both parameters significantly reduces the sample frequency, but leads to a clear degradation of regulation quality, as shown by the case $(2\eta_1, 2\eta_2)$. Therefore, η_1 and η_2 provide an explicit trade-off between sampling efficiency and closed-loop regulation performance, which is consistent with the discussion in Remark 4.

C. Analysis of the results under the proposed method

The state responses in Figure 3 show that the angular position and velocity converge to a small neighborhood of the origin under external disturbances, confirming the UUB properties established in Theorem 3 and Corollary 1. The sampled states closely track the actual states, while the phase portraits demonstrate effective disturbance attenuation and well-behaved transients under intermittent feedback.

TABLE V
SENSITIVITY ANALYSIS OF THE PARAMETERS η_1 AND η_2 .

(η_1, η_2) setting	Samples	Final state norm
(η_1^s, η_2^s)	51	0.000012
$(\eta_1^s, 0.5\eta_2^s)$	94	0.000011
$(\eta_1^s, 2\eta_2^s)$	30	0.00019
$(0.5\eta_1^s, \eta_2^s)$	85	0.000009
$(2\eta_1^s, \eta_2^s)$	39	0.00002
$(0.5\eta_1^s, 0.5\eta_2^s)$	162	0.000008
$(2\eta_1^s, 2\eta_2^s)$	25	0.011

Note: Here, $(\eta_1^s, \eta_2^s) = (0.43, 0.28)$ is specified in Table III.

As shown in Figure 4, the projected Actor updates in (32) maintain bounded control inputs without dense sampling or continuous state measurements. The zoomed-in results further indicate that the PO suppresses excessive oscillations and constraint violations under sparse updates.

In the NZSG setting, Figure 5 compares the cost performance of each player with and without the PO. As shown in Figure 5-a), the PO-based trajectories converge faster and settle at lower values, which is further confirmed by the bar chart in Figure 5-b). This indicates that the PO regularizes Actor updates and reduces the accumulated costs in this example. By bounding Actor weights, the PO implicitly constrains control inputs, suppresses excessive actions, and promotes stabilizing and efficient policies, thereby lowering J_i for all players.

Figure 6 evaluates policy-learning accuracy through the on-line errors of the Actor and Critic networks. The initially large deviations quickly decrease and remain bounded despite disturbances. These results are consistent with the ER-enhanced learning mechanism and illustrate effective approximation of Nash policies under the proposed hybrid update scheme.

Figure 7 further shows that the PO keeps Actor-weight norms within predefined bounds and prevents the parameter growth observed without projection. Together with Figure 5, these results confirm that PO improves stability and reduces the cost functions.

VIII. CONCLUSION

This paper proposes a novel fuzzy Actor-Critic RL algorithm for the multi-player differential NZSGs. By the presented self-triggered mechanism, the need for continuous state monitoring is eliminated, while excluding Zeno behavior. The integration of PO into the Actor update guarantees the boundedness of adaptive weights, which in turn results in lower cumulative cost functions and stability across all players. To further enhance learning efficiency, a composite learning strategy with ER is employed, enabling bounded approximation under SE condition, thus relaxing the need for PE. Theoretical analysis and simulations validate the the proposed approach, demonstrating robustness, stability, and reduced cost. Future research will extend the results to model-free settings.

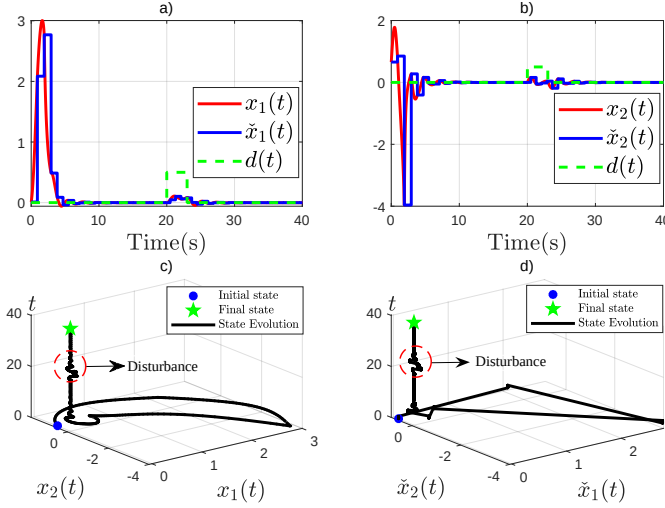


Fig. 3. The state trajectories under the proposed scheme. a) The angle. b) The angular velocity. c) The phase portrait with time evolution. d) The sampled phase portrait with time evolution.

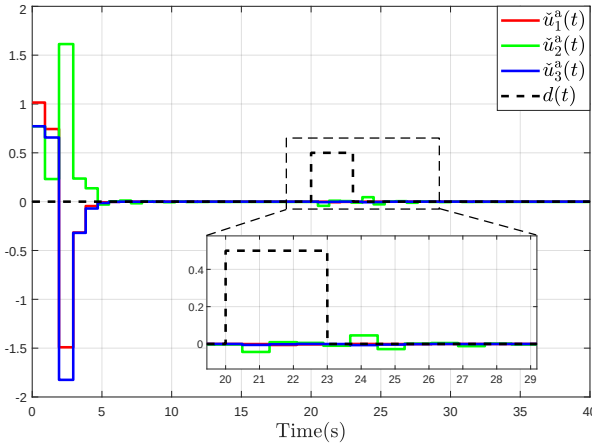


Fig. 4. The evolution of control inputs under unknown external disturbance.

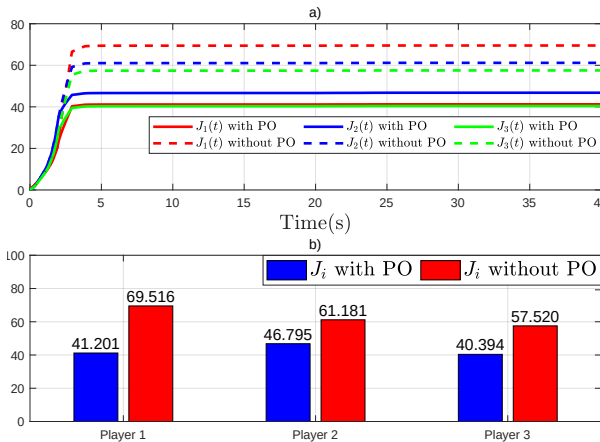


Fig. 5. The comparison of cost functions between with and without PO. a) The time evolution of each cost function J_i . b) The final value of each J_i .

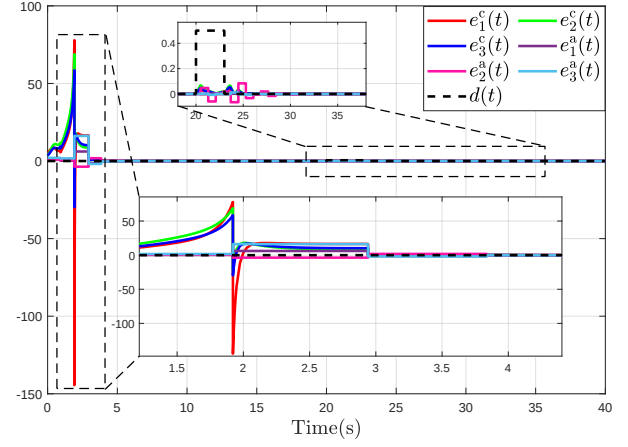


Fig. 6. The online learning errors of each Actor and Critic.

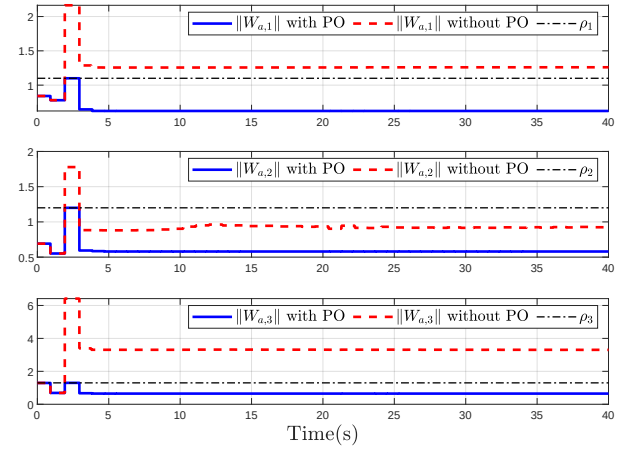


Fig. 7. The norms of Actor weights with and without PO.

APPENDIX A: PROOF OF THEOREM 2

Along the system (3) under the intermittent feedback (13):

$$\dot{x}(t) = f(x(t)) + \sum_i g_i(x(t)) \tilde{u}_i^*(t), \quad (49)$$

take the time derivative of $V_i^*(x(t))$ and bound it as follows:

$$\begin{aligned} \dot{V}_i^* &= \nabla^\top V_i^* \cdot (f + \sum_j g_j u_j^*) \\ &= \nabla^\top V_i^* \sum_j g_j (\tilde{u}_j^* - u_j^*) + \nabla^\top V_i^* (f + \sum_j g_j u_j^*) \\ &\leq \|\nabla V_i^*\| \cdot \sum_j \|g_j\| \cdot |\tilde{u}_j^* - u_j^*| + \nabla^\top V_i^* (f + \sum_j g_j u_j^*), \end{aligned}$$

where the first term can be further bounded via $\|\nabla V_i^*\| \leq \delta_i$ (i.e., (11)), $\|g_j\| \leq \mathcal{G}_j$ and the Lipschitzness (10) of u_i^* as

$$\|\nabla V_i^*(x) \cdot \sum_j \|g_j(x)\| \cdot |u_j^*(\tilde{x}) - u_j^*(x)| \leq \delta_i \cdot \sum_j \mathcal{G}_j \mathcal{L}_j \|\tilde{x}\|,$$

and the second term $\nabla^\top V_i^* \cdot (f + \sum_j g_j u_j^*) = -c_i^*$ by the Bellman equation (7). Therefore,

$$\dot{V}_i^*(t) \leq -c_i^*(x(t)) + \delta_i \cdot \sum_j \mathcal{G}_j \mathcal{L}_j \|\tilde{x}(t)\|.$$

Since $\sum_j r_{ij} u_j^*(x)^2 \geq 0$ and $Q_i(x) \geq \underline{q}_i \|x\|^2$ (i.e., (5)),

$$c_i^*(x) = Q_i(x) + \sum_j r_{ij} u_j^*(x)^2 \geq \underline{q}_i \|x\|^2$$

by which $\dot{V}_i^*(t) \leq -\underline{q}_i \|x(t)\|^2 + \delta_i \cdot \sum_j \mathcal{G}_j \mathcal{L}_j \|\tilde{x}(t)\|$.

Now, consider $V(x) \triangleq \sum_i V_i^*(x)$ as a Lyapunov candidate whose derivative along the system (49) satisfies

$$\dot{V}(x) = \sum_i \dot{V}_i^*(x) \leq -\underline{q}\|x(t)\|^2 + \delta \cdot \mathcal{G}_{\mathcal{L}}\|\tilde{x}(t)\|$$

where $\underline{q} \triangleq \sum_i \underline{q}_i$, $\delta \triangleq \sum_i \delta_i$ and $\mathcal{G}_{\mathcal{L}} \triangleq \sum_j \mathcal{G}_j \mathcal{L}_j$. Moreover, V is positive definite as so is each V_i^* by the ZSO condition, and the application of Lemma 5 further yields

$$\begin{aligned} \dot{V}(x) &\leq -\underline{q}\|x\|^2 + \delta \cdot \mathcal{G}_{\mathcal{L}} \cdot \sqrt{\eta_1\|x\|^2 + \eta_1\eta_2} \\ &= -\beta' \underline{q}\|x\|^2 - \left[\beta \underline{q}\|x\|^2 - \delta \mathcal{G}_{\mathcal{L}} \cdot \sqrt{\eta_1\|x\|^2 + \eta_1\eta_2} \right], \end{aligned}$$

where $\beta' \triangleq 1 - \beta$ and $\beta \in (0, 1)$. The term in the brackets is non-negative iff

$$(\beta \underline{q})^2 \|x\|^4 \geq (\delta \mathcal{G}_{\mathcal{L}})^2 \cdot (\eta_1 \|x\|^2 + \eta_1 \eta_2), \quad (50)$$

which is quadratic in $\|x\|^2$ and true iff $\|x\| \geq \hat{x}$, where

$$\hat{x} \triangleq \frac{\delta \mathcal{G}_{\mathcal{L}}}{\beta \underline{q}} \sqrt{\frac{\eta_1 + \sqrt{\eta_1^2 + 4\eta_1\eta_2(\beta \underline{q}/\delta \mathcal{G}_{\mathcal{L}})^2}}{2}}$$

can be made arbitrary small by choosing the parameters η_1 and η_2 small enough. I.e., $\dot{V}(x) \leq -\beta' \underline{q}\|x\|^2$ if $\|x\| \geq \hat{x}$. Additionally, it follows from Assumption 2 that

$$\underline{\alpha}(\|x\|) \triangleq \sum_i \underline{\alpha}_i(\|x\|) \leq V(x) \leq \sum_i \bar{\alpha}_i(\|x\|) \triangleq \bar{\alpha}(\|x\|). \quad (51)$$

Therefore, $x(t)$ is UUB [36, Theorem 4.18], provided that η_1 and η_2 are small enough, so that the neighborhood of the origin $\{x \in \mathbb{R}^n : \|x\| \leq \underline{\alpha}^{-1}(\bar{\alpha}(\hat{x}))\}$ is contained by Ω .

APPENDIX B: PROOFS OF THEOREM 3 AND COROLLARY 1

Proof of Theorem 3: To begin with, recall the Lyapunov inequality (36) and let $V = \sum_i V_i^*$ (see Appendix A), whose time derivative is upper bounded as $\dot{V}(x) \leq -\underline{q}\|x\|^2 + \delta \cdot \mathcal{G}_{\mathcal{L}}\|\tilde{x}\| + \delta \cdot \mathcal{E}$, where $\underline{q} = \sum_i \underline{q}_i$ and $\delta = \sum_i \delta_i$. Applying Lemma 5, we obtain

$$\begin{aligned} \dot{V}(x) &\leq -\kappa_1 \underline{q}\|x\|^2 - (\kappa_2 \underline{q}\|x\|^2 - \delta \mathcal{G}_{\mathcal{L}} \sqrt{\eta_1\|x\|^2 + \eta_1\eta_2}) \\ &\quad - (\kappa_3 \underline{q}\|x\|^2 - \delta \mathcal{E}) \end{aligned} \quad (52)$$

where κ_1, κ_2 and κ_3 are constants in $(0, 1)$ s.t. $\kappa_1 + \kappa_2 + \kappa_3 = 1$. Accordingly, $\dot{V} \leq -\kappa_1 \underline{q}\|x\|^2$, provided that $\|x\| \geq \underline{x}$ for

$$\underline{x} \triangleq \max \left\{ \sqrt{\frac{\delta \mathcal{E}}{\kappa_3 \underline{q}}}, \frac{\delta \mathcal{G}_{\mathcal{L}}}{\kappa_2 \underline{q}} \sqrt{\frac{\eta_1 + \sqrt{\eta_1^2 + 4\eta_1\eta_2(\kappa_2 \underline{q}/\delta \mathcal{G}_{\mathcal{L}})^2}}{2}} \right\}.$$

Furthermore, under (51), by the UUB Theorem [36, Theorem 4.18], there exists a finite time $\mathbf{T}_x$ s.t.

$$\|x(t)\| \leq \bar{x} \triangleq \underline{\alpha}^{-1}(\bar{\alpha}(\underline{x})) \quad \text{for } t \geq \mathbf{T}_x, \quad (53)$$

which shows the UUB property of the state $x(t)$. Moreover, smaller ρ_i (i.e., smaller $\mathcal{E}$, see (36)) together with smaller η_1 and η_2 result in a tighter bound $\bar{x}$ on the system state $\|x\|$.

Proof of Corollary 1: Next, we derive the explicit bound of the sampled state using (38). Recall (52). With $v > 0$, and the Young's inequality: $\delta \mathcal{G}_{\mathcal{L}} \sqrt{\eta_1\|x\|^2 + \eta_1\eta_2} \leq v(\delta \mathcal{G}_{\mathcal{L}})^2/2 + (\eta_1\|x\|^2 + \eta_1\eta_2)/2v$, we further get $\dot{V}(x) \leq -\kappa V(x) + \bar{\delta}_x$,

where $\kappa \triangleq (q - \frac{\eta_1}{2v})/\bar{\theta}$ and $\bar{\delta}_x \triangleq v(\delta \mathcal{G}_{\mathcal{L}})^2/2 + \eta_1\eta_2/(2v) + \delta_x$. During the execution interval $t \in [t_k, t_{k+1})$, by the Comparison Lemma [36], one can obtain the following inequality:

$$V(x(t)) \leq (V(x(t_k)) - \bar{\delta}_x/\kappa)e^{-\kappa(t-t_k)} + \bar{\delta}_x/\kappa. \quad (54)$$

At the update instant $t = t_{k+1}$, (54) implies

$$\begin{aligned} \Delta V(x(t_k)) &\leq -(1 - e^{-\kappa \Delta t_k})V(x(t_k)) + \bar{\delta}_x(1 - e^{-\kappa \Delta t_k})/\kappa \\ &\leq -\delta_{\tilde{x}} \underline{\theta} \|\tilde{x}\|^2 + \bar{\delta}_x \kappa \delta_{\tilde{x}}/\kappa, \end{aligned} \quad (55)$$

where $\delta_{\tilde{x}} \triangleq 1 - e^{-\kappa \Delta t_k}$. By the discrete-time UUB theorem [39], one can ensure that there exist constants $N_{\tilde{x}} \in \mathbb{N}$ and $\varsigma_{\tilde{x}} \in (0, 1)$ s.t. $\|\tilde{x}(t_k)\| \leq \bar{\tilde{x}} \triangleq \sqrt{\bar{\theta} \delta_{\tilde{x}}/\underline{\theta}^2 \kappa \varsigma_{\tilde{x}}} \quad \forall k \geq N_{\tilde{x}}$, which completes the proof.

APPENDIX C: PROOF OF THEOREM 4

First, for the continuous Critic network, recall (46), in which the term $-\tilde{W}_i^{\text{c}\top} \tilde{\omega}_i \tilde{\omega}_i^{\top} \tilde{W}_i^{\text{c}}$ is non-positive. Additionally, since the signal $\varpi_{i\ell}$ is SE, i.e., $\lambda(\mathcal{C}_{i1}) \geq \beta_i^{\text{c}}$ for some constant $\beta_i^{\text{c}} > 0$, the term $-\tilde{W}_i^{\text{c}\top} \mathcal{C}_{i1} \tilde{W}_i^{\text{c}}$ satisfies

$$-\tilde{W}_i^{\text{c}\top} \mathcal{C}_{i1} \tilde{W}_i^{\text{c}} \leq -\beta_i^{\text{c}} \|\tilde{W}_i^{\text{c}}\|^2.$$

Then, it remains to bound the term $\tilde{W}_i^{\text{c}\top} \mathcal{C}_{i2}$ that contains the residual ζ_i^{c} (see (43) and (44)). First, we derive the bounds of γ_{ih} contained in the expression of ζ_i^{c} . Under Assumption 3 and Lemma 2, the “ u -terms” in γ_{ih} are bounded as

$$\begin{aligned} \|\tilde{u}_j^{\text{a}}(t)\| &= \|W_j^{\text{a}\top} \tilde{\psi}_j^{\text{a}}\| \leq \tilde{\mathcal{U}}_j \triangleq \rho_j \bar{\psi}^{\text{a}}, \\ \|\tilde{u}_j^{\text{a}\star}(t)\| &= \|W_j^{\text{a}\star\top} \tilde{\psi}_j^{\text{a}} + \tilde{\varepsilon}_j^{\text{a}}\| \leq \tilde{\mathcal{U}}_j^{\star} \triangleq \|W_j^{\text{a}\star}\| \bar{\psi}^{\text{a}} + \bar{\varepsilon}^{\text{a}}, \\ \|\tilde{u}_j^{\text{a}}(t)\| &= \|\tilde{W}_j^{\text{a}\top} \tilde{\psi}_j^{\text{a}} + \tilde{\varepsilon}_j^{\text{a}}\| \leq \tilde{\mathcal{U}}_j \triangleq (\|W_j^{\text{a}\star}\| + \rho_j) \bar{\psi}^{\text{a}} + \bar{\varepsilon}^{\text{a}}, \end{aligned}$$

which further contributes to the boundedness of $\gamma_{i,h}$ s.t.

$$\begin{aligned} \gamma_{i1} &\leq \bar{\gamma}_{i1} \triangleq \sum_j r_{ij} (\tilde{\mathcal{U}}_j^2 + \tilde{\mathcal{U}}_j^{\star 2}), \quad \gamma_{i2} \leq \bar{\gamma}_{i2} \triangleq \bar{\varepsilon}_d^{\text{c}} \sum_j \mathcal{G}_j \tilde{\mathcal{U}}_j^{\star}, \\ \gamma_{i3} &\leq \bar{\gamma}_{i3} \triangleq \|W_i^{\text{c}\star}\| \bar{\psi}_d^{\text{c}} \sum_j \mathcal{G}_j (2\tilde{\mathcal{U}}_j + \tilde{\mathcal{U}}_j^{\star}). \end{aligned}$$

Then, based on above bounds and (53), the Critic residual ζ_i^{c} is bounded by $\zeta_i^{\text{c}}(t) \leq \bar{\zeta}_i^{\text{c}} = \bar{\varepsilon}_d^{\text{c}} \mathcal{L}_f \bar{x} + \sum_{h=1}^3 \bar{\gamma}_{ih}$, for $t \geq \mathbf{T}_x$, where $\bar{x}$ and $\mathbf{T}_x$ are defined in (53).

Therefore, with the fact that $\frac{a}{1+a^2} \leq \frac{1}{2}$ for all $a \in \mathbb{R}$, one has $\|\mathcal{C}_{i2}\| \leq \bar{\mathcal{C}}_{i2} \triangleq (1 + L_i^{\text{c}}) \bar{\zeta}_i^{\text{c}}/2$. By the inequalities set forth, (46) can be further bounded as

$$\dot{W}_i^{\text{c}} \leq -\beta_i^{\text{c}} \|\tilde{W}_i^{\text{c}}\|^2 + \|\tilde{W}_i^{\text{c}}\| \bar{\mathcal{C}}_{i2}. \quad (56)$$

By the UUB theorem [36, Theorem 4.18], there exists $\varsigma_i^{\text{c}} \in (0, 1)$ and a finite time $\mathbf{T}_i^{\text{c}} > \mathbf{T}_x$ s.t.

$$\|\tilde{W}_i^{\text{c}}\| \leq \tilde{W}_{i,\max}^{\text{c}} \triangleq \bar{\mathcal{C}}_{i2}/(\beta_i^{\text{c}} \varsigma_i^{\text{c}}) \quad \text{for } t \geq \mathbf{T}_i^{\text{c}}. \quad (57)$$

Next, we turn our attention to the discrete Actor weight errors $\tilde{W}_i^{\text{a}}$ but consider the trigger instants $t = t_{k+1} > \mathbf{T}_i^{\text{c}}$ only, for all i . This implies (53) and (57) hold. For ΔV_i^{a} , recall (41) that includes the “ Ξ -terms” in (42) and “ $\mathcal{A}$ -terms” in (40). First, considering the “ $\mathcal{A}$ -terms” containing ζ_i^{a} , one has by Assumption 3 that

$$\zeta_i^{\text{a}} \leq \|W_i^{\text{a}\star}\| \bar{\psi}^{\text{a}} + \mathcal{G}_i \bar{\psi}_d^{\text{c}} \|W_i^{\text{c}}\|/(2r_{ii}) \leq \bar{\zeta}_i^{\text{a}}$$

where $\bar{\zeta}_i^a \triangleq \|W_i^{a*}\|\bar{\psi}^a + \frac{1}{2r_{ii}}\mathcal{G}_i\bar{\psi}^c(\|W_i^{c*}\| + \tilde{W}_{i,\max}^c)$. Hence, $\mathcal{A}_{i2}$ and $\mathcal{A}_{i4}$ satisfy $\|\mathcal{A}_{i2}\| \leq \bar{\mathcal{A}}_{i2}$ and $\|\mathcal{A}_{i4}\| \leq \bar{\mathcal{A}}_{i4}$, where

$$\bar{\mathcal{A}}_{i2} \triangleq (1 + L_i^a)\bar{\psi}^a\bar{\zeta}_i^a, \\ \bar{\mathcal{A}}_{i4} \triangleq \alpha_i^a\bar{\psi}^a(r_{ii}^{-1}\mathcal{G}_i\bar{\psi}^c\|W_i^{c*}\| + r_{ii}^{-1}\mathcal{G}_i\bar{\varepsilon}^c + 2\bar{\varepsilon}^a + 2\bar{\psi}^a\|W_i^{a*}\|)/2.$$

Therefore, one can obtain the bounds of Ξ_{i1} and Ξ_{i2} as

$$\begin{aligned} \Xi_{i1} &\leq (\alpha_i^a)^2(\bar{\mathcal{A}}_{i2})^2 + (\bar{\mathcal{A}}_{i4})^2 + 2\alpha_i^a\bar{\mathcal{A}}_{i2}\bar{\mathcal{A}}_{i4} \triangleq \bar{\Xi}_{i1}, \\ \|\Xi_{i2}\| &\leq 2[\alpha_i^a\bar{\mathcal{A}}_{i2} + \bar{\mathcal{A}}_{i4} + (\alpha_i^a)^2\lambda_i^a\bar{\mathcal{A}}_{i2} + \alpha_i^a\lambda_i^a\bar{\mathcal{A}}_{i4}] \\ &= 2[\alpha_i^a\bar{\mathcal{A}}_{i2}(1 + \alpha_i^a\lambda_i^a) + \bar{\mathcal{A}}_{i4}(1 + \alpha_i^a\lambda_i^a)] \\ &= 2(\alpha_i^a\bar{\mathcal{A}}_{i2} + \bar{\mathcal{A}}_{i4})(1 + \alpha_i^a\lambda_i^a) \triangleq \bar{\Xi}_{i2}, \end{aligned} \quad (58)$$

where $\lambda_i^a \triangleq \bar{\lambda}(\bar{\psi}_i^a\bar{\psi}_i^{a\top} + \mathcal{A}_{i1})$. Additionally, Ξ_{i3} is always non-positive as follows.

- 1) If $\|\tilde{W}_i^a\| \leq \rho_i$, then $\mathcal{A}_{i3} = 0$ by definition $\Rightarrow \Xi_{i3} = 0$.
- 2) If $\|\tilde{W}_i^a\| > \rho_i$, then $\mathcal{A}_{i3} = \widehat{W}_i^a - \bar{W}_i^a$ for $\widehat{W}_i^a = \rho_i \tilde{W}_i^a / \|\tilde{W}_i^a\|$ indicating that Ξ_{i3} can be bounded as

$$\begin{aligned} \Xi_{i3} &= \|\mathcal{A}_{i3}\|^2 - 2\mathcal{A}_{i3}^\top(\bar{W}_i^{a*} - \widehat{W}_i^a) - 2\mathcal{A}_{i3}^\top(\widehat{W}_i^a - \bar{W}_i^a) \\ &= -\|\mathcal{A}_{i3}\|^2 - 2(\widehat{W}_i^a - \bar{W}_i^a)^\top(\bar{W}_i^{a*} - \widehat{W}_i^a) \leq 0. \end{aligned}$$

≥ 0 by Lemma 2

Then, with the above facts, (58), and $\beta_i^a \leq \lambda(\Lambda_{i1}^a)$ for some constant $\beta_i^a > 0$ by the SE condition, ΔV_i^a in (41) satisfies

$$\Delta V_i^a \leq -(2\alpha_i^a\beta_i^a - (\alpha_i^a)^2(\lambda_i^a)^2)\|\tilde{W}_i^a\|^2 + \bar{\Xi}_{i1} + \|\tilde{W}_i^a\|\bar{\Xi}_{i2},$$

Applying Young's inequality to the term $\|\tilde{W}_i^a\|\bar{\Xi}_{i2}$, one has

$$\|\tilde{W}_i^a\| \cdot \bar{\Xi}_{i2} \leq v_i^a\|\tilde{W}_i^a\|^2/2 + \bar{\Xi}_{i2}^2/(2v_i^a)$$

for some $v_i^a > 0$, which implies $\Delta V_i^a \leq -\mathfrak{L}(\alpha_i^a)\|\tilde{W}_i^a\|^2 + \bar{\Xi}_{i1} + \frac{\bar{\Xi}_{i2}^2}{2v_i^a}$, where $\mathfrak{L}(\alpha_i^a) \triangleq 2\alpha_i^a\beta_i^a - (\alpha_i^a)^2(\lambda_i^a)^2 - \frac{v_i^a}{2}$. Choose $v_i^a \leq 2\beta_i^a/(\lambda_i^a)^2$ and $\alpha_i^a \in (\alpha_i^{a-}, \alpha_i^{a+}) \subset (0, 2\beta_i^a(\lambda_i^a)^2)$, where $\alpha_i^{a\pm} \triangleq (\beta_i^a \pm \sqrt{(\beta_i^a)^2 - v_i^a(\lambda_i^a)^2/2})/(\lambda_i^a)^2$. This implies $\mathfrak{L}(\alpha_i^a) > 0$. Hence, by the discrete-time UUB theorem in [39, Corollary 2.5], $\exists N_i^a \in \mathbb{N}^+$ and $\varsigma_i^a \in (0, 1)$ s.t.

$$\|\tilde{W}_i^a(t_k)\| \leq \sqrt{\frac{2v_i^a\bar{\Xi}_{i1} + (\bar{\Xi}_{i2})^2}{2v_i^a\varsigma_i^a\mathfrak{L}(\alpha_i^a)}} \quad \text{for } k \geq N_i^a.$$

This completes the proof.

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